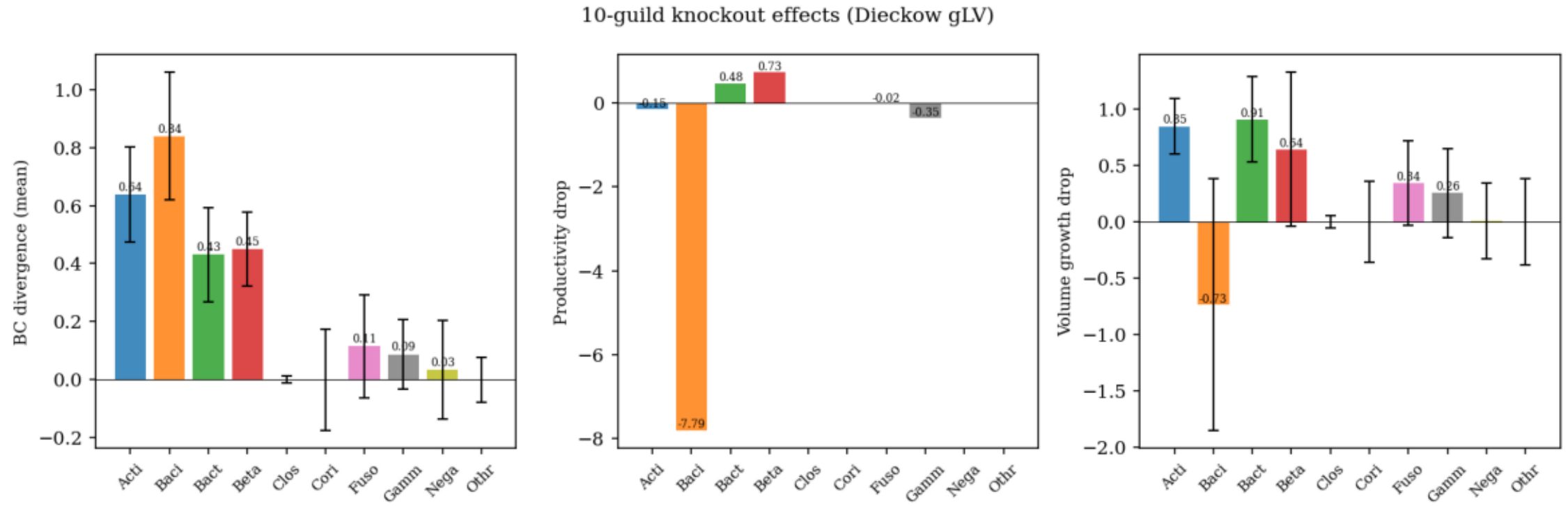


Guild-Level Dynamical Analysis

Keystone · Network · Tipping · Relapse · Prevention · Phase Diagram

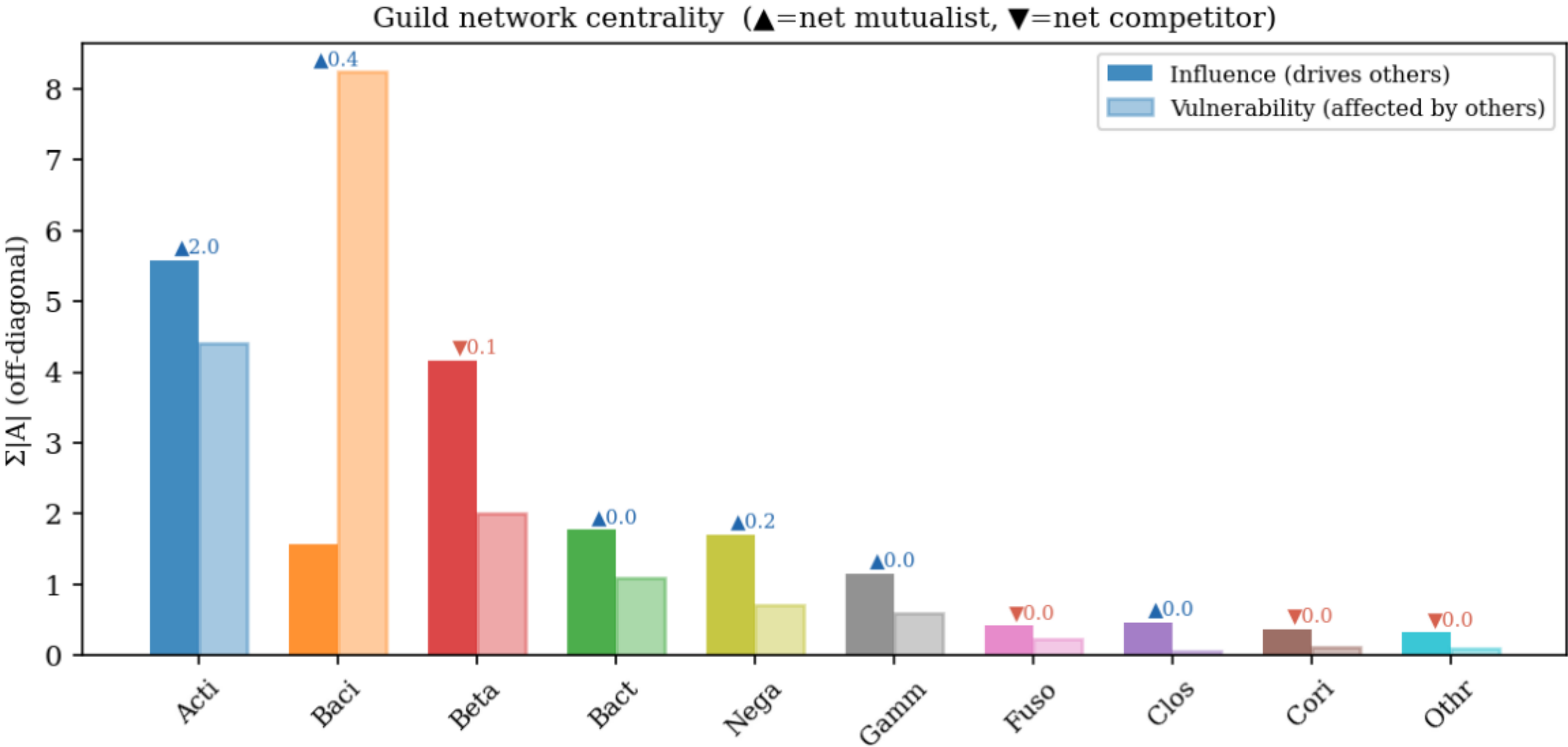
Dieckow 2024 · 10-guild gLV · N=10 patients · Nishioka (2026)

Which guild, when removed, shifts the community most?



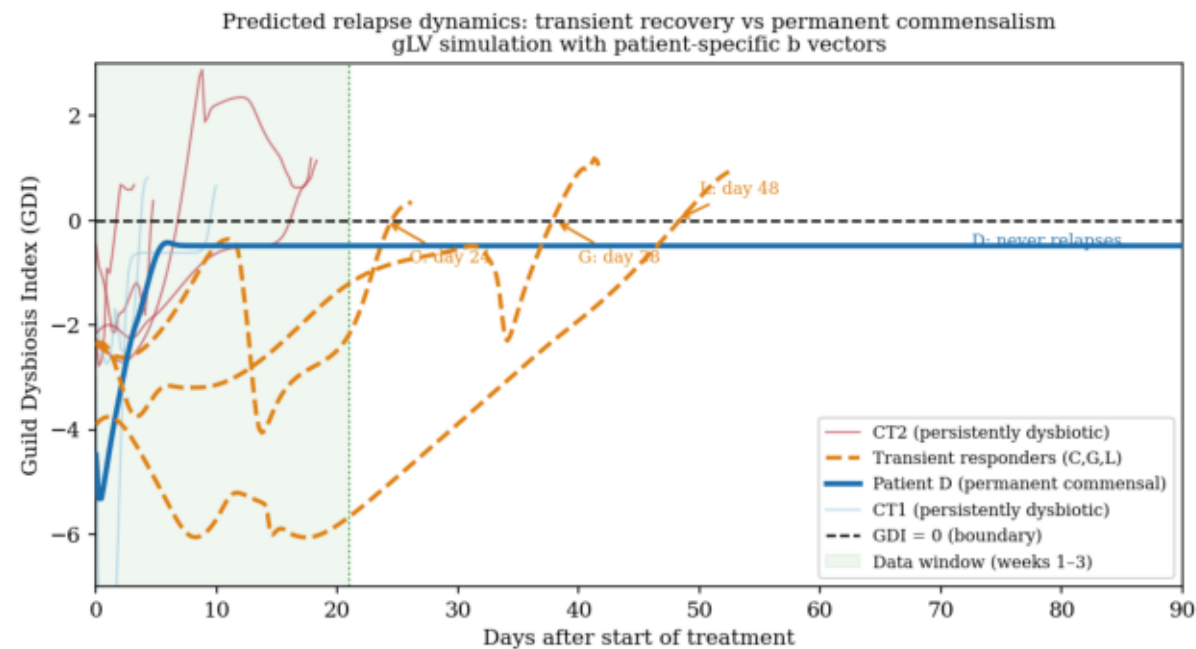
Bacilli: highest BC divergence (0.839) — compositional keystone. Removal increases volume → competitive suppressor. | Actinobacteria: vol_drop = +0.846 → growth engine. Strongest single interaction: Acti → Baci A = +3.43.

Column-sum $|A|$ = influence | Row-sum $|A|$ = vulnerability | $\sum A_{ij}$ = net mutualist (+) / competitor (−)

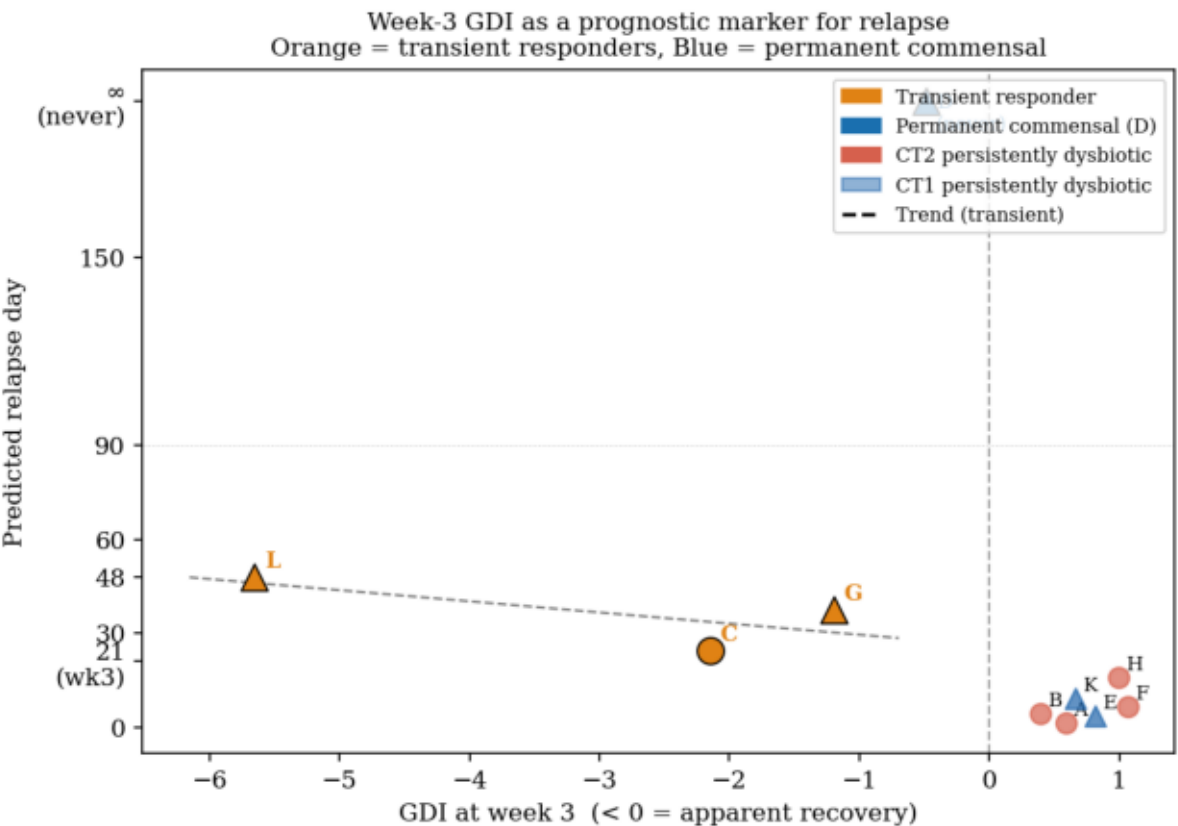


Actinobacteria: influence = 5.58 (highest), net = +2.02 (mutualist hub). Bacilli: vulnerability = 8.24 (most regulated). Betaproteobacteria: net = −0.14 → net competitor. A matrix encodes stable dysbiotic attractor via Acti–Baci–Beta triad.

Long-time simulation (t = 150 d) from patient-specific b vectors | week-3 GDI as prognostic marker



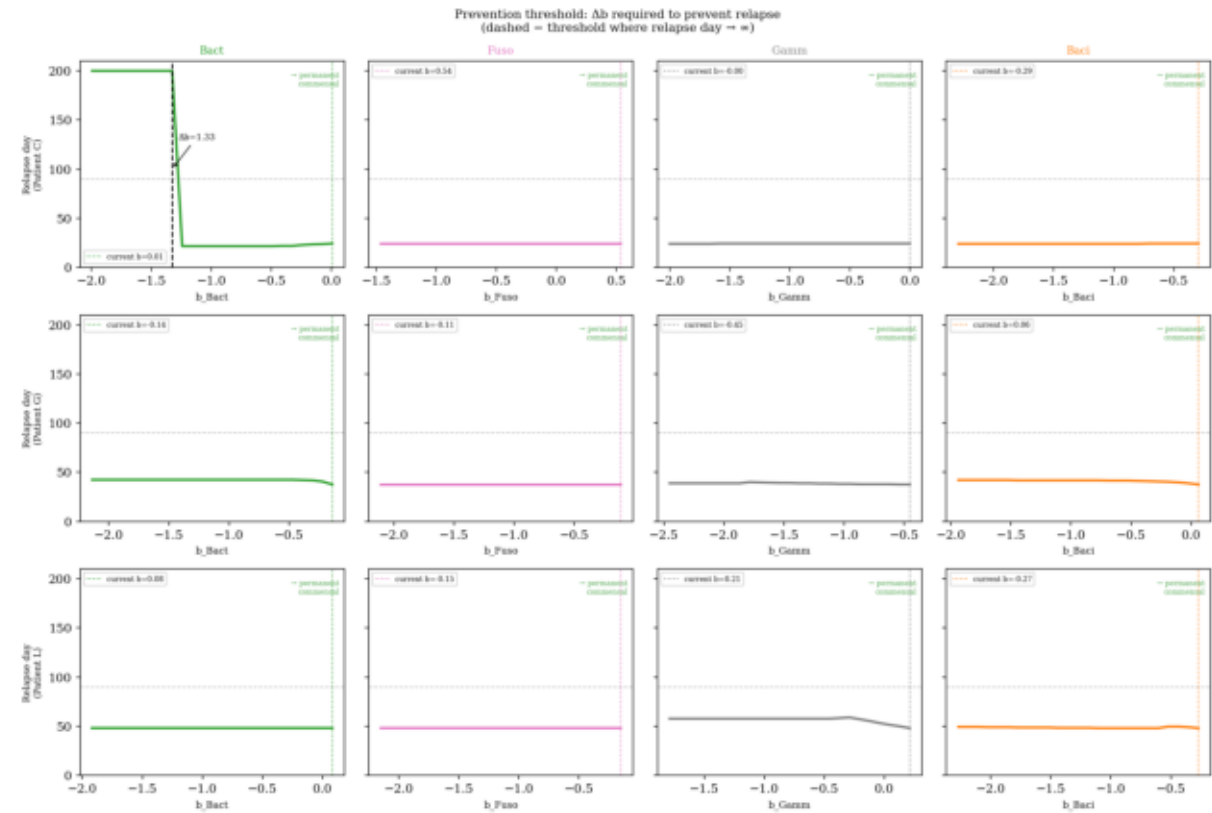
GDI trajectories to 90 days (per patient)



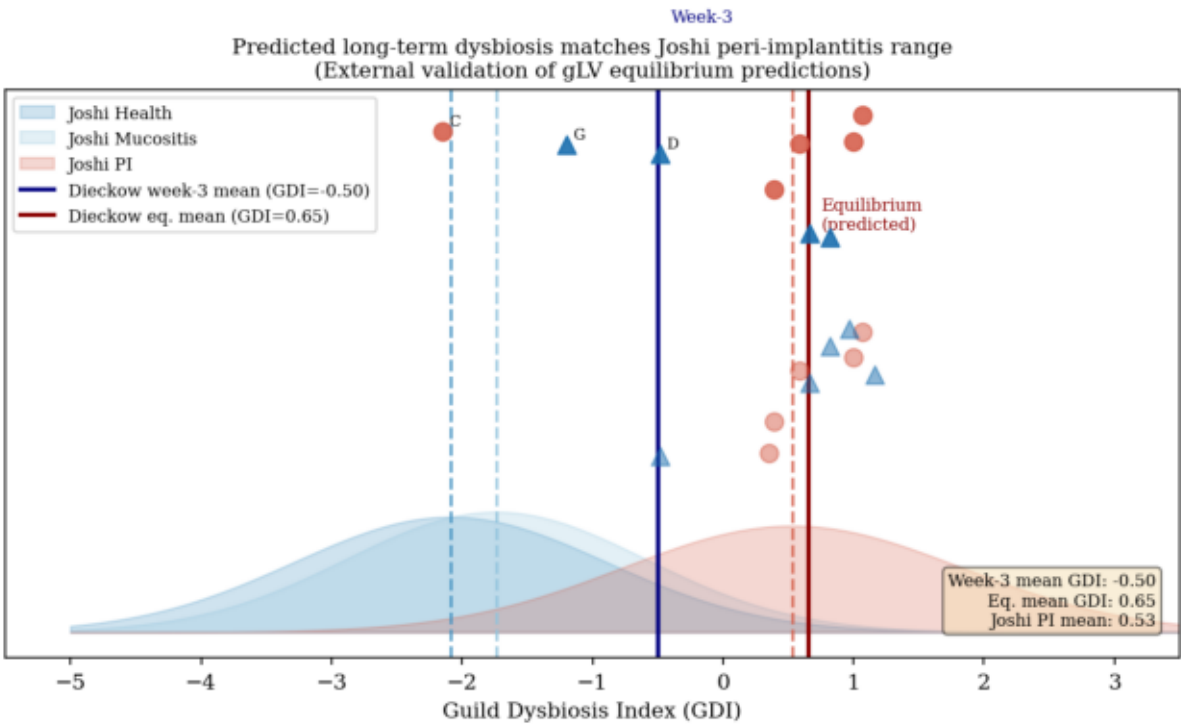
Week-3 GDI → relapse day

① Persistent dysbiotic (A,B,E,F,H,K) ② Transient responder: C→24d, G→38d, L→48d (commensal at wk-3, dysbiotic by t<50d) ③ Permanent commensal (D only). → Lower week-3 GDI = longer remission. 3-week data extrapolated to 150d.

Min. Δb_i to prevent relapse | Model equilibrium GDI vs peri-implantitis reference



Required Δb per guild (patient C only)

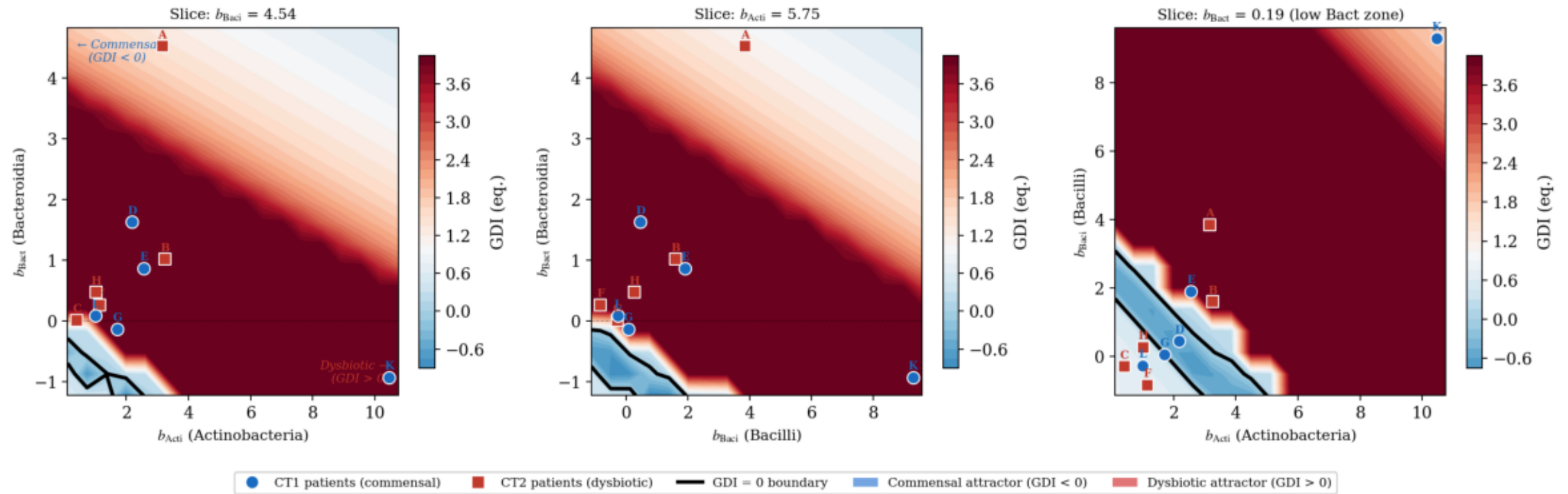


Predicted eq. GDI vs Joshi PI distribution

Prevention: only Patient C convertible by single-guild intervention ($\Delta b_{\text{Bact}} = 1.33$). G & L require multi-guild change $\rightarrow$ deep attractor. | Joshi validation: predicted eq. GDI = +0.65 $\approx$ Joshi PI mean +0.53 (9/10 patients in PI range, difference $< 0.12 < \text{Joshi SD}=1.30$).

Equilibrium GDI on 18^3 grid ($b_{Acti} \times b_{Baci} \times b_{Bact}$), $t = 80$ d, 12-core parallel. Blue = commensal | Red = dysbiotic | Black contour = GDI = 0

gLV attractor landscape: equilibrium GDI as a function of guild-specific growth rates
(other b_i fixed at population mean; colour saturated at $GDI \in [-2, +4]$)



98% of parameter space is dysbiotic. Commensal region confined to $b_{Bact} < \sim 1.5$ (Bacteroidia growth rate is the dominant control variable). CT1 patients with low b_{Bact} cluster near the boundary; CT2 patient A ($b_{Bact}=4.5$) is deep in the dysbiotic zone. → Structural attractor robustness — not initial-condition sensitive.